CASE STUDY ANALYSIS

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TASK RUNDOWN

- Dataset A – 15 000 Insurance Cases for Training
- Dataset B – 3 000 Insurance Cases for Prediction (newer data, no target column)
- Data Description
- A Premade „risk scorer“ model

EXPLORATORY ANALYSIS

- Model split into AL & APD cases
- Vehicles are insured in fleets
  - Not a problem, but may result in biased data entering our model
- Variable names for identical values (CAPSED variants in business type)
- Dataset includes missing values
- Inclusion of duplicate rows
  - Identical
  - Non-Identical (capitalization of business_type column)
- Score dataset is missing the **“**has_safety_program**”** column
- 10% of score rows have post-claim columns filled in, which shouldnt exist at policy start
- Sanity checks

MODEL FIXES

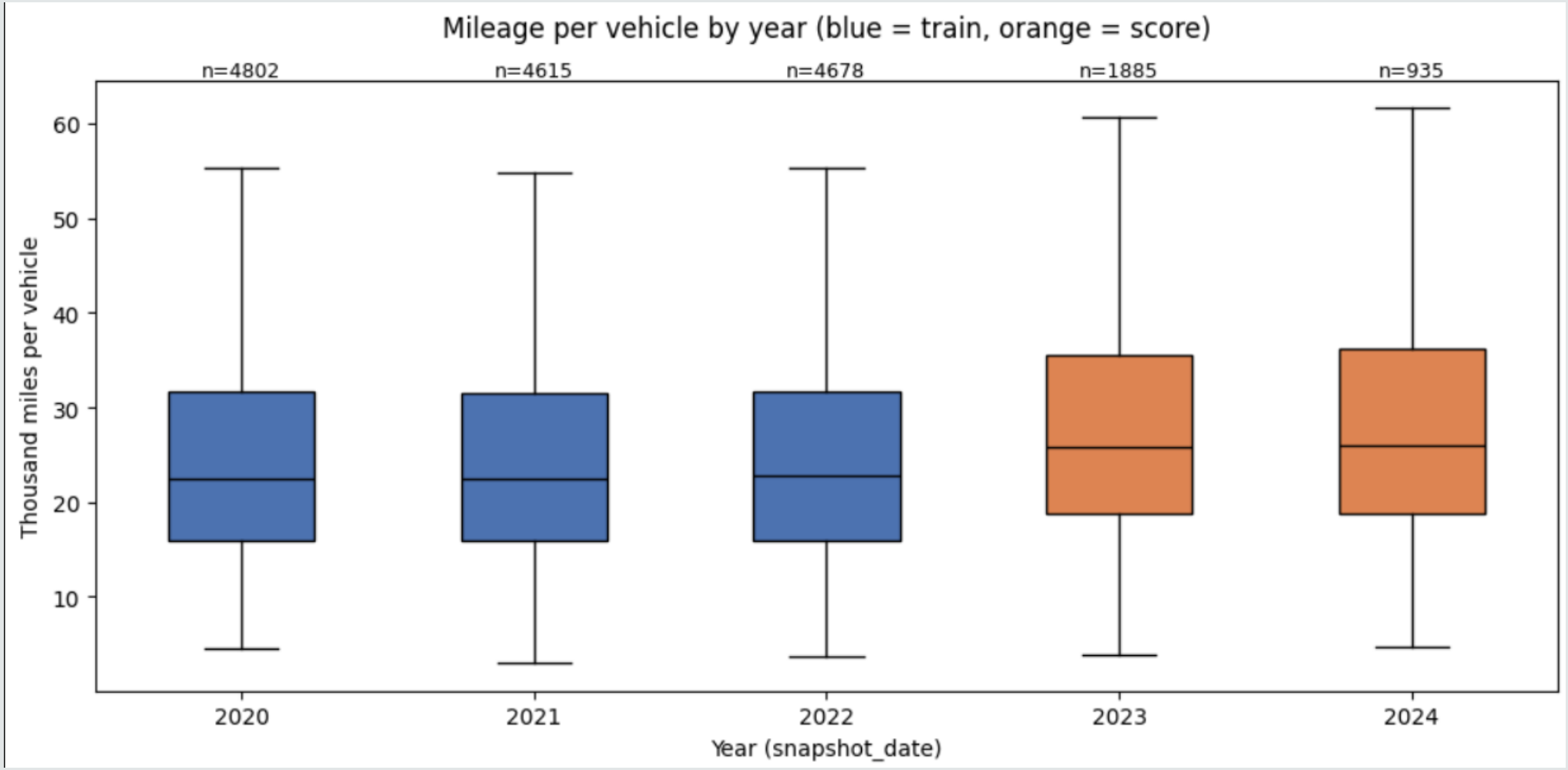
- Data leakage (claim_paid_amount_current_period, days_to_first_claim_report)
  - Potential leakage in "late_payment_count“ and risk_score_external
- Filling all missing values with 0 creates impossible values (driver_avg_age, years_in_business)
  - I decided to go with a simpler median filler combined with a missing indicator flag, but switching to a gradient boosting model might be more reliable
- Possible problems with encoding datasets separately (fixed by creating a pipeline)
- The model is scored on predicting its training data

BEYOND IMPROVEMENTS

- Added a fixed seed to create reproducible results
- Final model refit on all data
- Different modeling choices:
  - The dataset forms are not very compatible with a logistic regression binary classifier model
  - Would recommend risk-level predicting model
  - APD and AL might benefit from having two separate models
- Overall prediction might be biased due to being recorded in different years

- Model alternatives I would consider
  - Random forests, XGBoost, **HistBoosting**
  - (mainly due to possible nonlinear dependencies)

WHY OUT OF TIME VALIDATION COULD MATTER



TOP 3 MODELING IMPROVEMENTS

- Including a out of time validation
  - Training on 2020-21 and validate on 2022
- Setting a better target
  - Expected loss instead of “had a claim”
- Splitting the model into two
  - Separating models for AL and APD

TOP 3 PRODUCTION READINESS IMPROVEMENTS

- Input data validation
  - This would fix missing columns, differences in data inputs
- Turn the notebook into modular code
  - Separate .py files for training, scoring
  - A config file with feature list, seed, ..
  - Maybe some unit tests
- Logging, monitoring, versioning
  - Monitor actual performance, stamp data with the model version

DEPLOYMENT ROADMAP

- Validate with business and create in-detail documentation
  - Check on the validity of the success metric
- Integrate
- Real use-case test phase
  - See the model performance in real scenarios before relying on it for decisions
- Deploy with monitoring dashboards, alerts, ..

WHAT I LEARNED

- I went deep on trying to find a better suited model, which has led to furthering my understanding of different variants
- I learned about different missing value filling methods

AI TOOLS USED

- All uses of generative AI are disclosed in the final notebook (used for making the coding process faster)
- I have also used AI to check for misspellings and logical errors in sentences

LLM Used: Claude Opus 5.5 (medium effort)

SUMMARIZING MY EXPERIENCE

- Overall I liked this task even as just an excercise, because it made me try and look at the data given more rigorously
- I liked catching all the little planted bugs and mistakes